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(HW-3)

40.570 Stochastic Models.

(1) If X is exponential with rate λ.

RTP: $Y = \lfloor X \rfloor + 1$ is geometric with param $p = 1 - e^{-\lambda}$ where $\lfloor x \rfloor$ is floor func^n.

For $n = 1, 2 \ldots$

$X \sim Exp(\lambda)$

$\therefore \boxed{P(Y = n) = P(n-1 \le X < n).}$ $\to$ because of floor function

Using the exponential cdf: $\boxed{\because P(Y=n) = P(X \geq n-1) - P(X \geq n)}$

$P(Y = n) = e^{-\lambda(n-1)} \cdot [1 - e^{-\lambda}].$

Now, if we set $p = 1 - e^{-\lambda}$

$\Rightarrow e^{-\lambda} = 1 - p.$

$\therefore P(Y = n) = (1-p)^{n-1} p. \Rightarrow \underbrace{p(1-p)^{n-1}}_{\text{Geometric dist.}}$

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(2) Stopping conditions.

(*) Poisson process on $[0, T]$, stop exactly at last event before T.

(*) For stopping time $t \in (s, T]$, winning condition is if and ONLY if there are NO events after that until T.

(*) Therefore stopping event must be: (i) first event after s. (ii) last event before T.

$\therefore$ Poisson increments:

$N(T) - N(S) \sim \text{Poisson}[\lambda(T-S)]$.

$\therefore P(\text{win}) = P(N(T) - N(S) = 1)$

(i) $\therefore P(N=K) = \dfrac{[\lambda(T-S)]^K}{K!} e^{-[\lambda(T-S)]}$.

For $K = 1$.

$P(\text{win}) = \boxed{\lambda(T-S)\, e^{-\lambda(T-S)}}$ (ans).

(ii) $\dfrac{\partial P}{\partial S} = \lambda(T-S)\left\{ e^{-\lambda(T-S)} [-\lambda(-1)] \right\} + \lambda(-1)\, e^{-\lambda(T-S)} = 0$.

Extremizing

$\underbrace{e^{-\lambda(T-S)}}\left[(T-S)\lambda - 1\right] = 0$.

Admits undefined solutions.

$\therefore (T-S)\lambda - 1 = 0$

$(T-S)\lambda = 1$.

$\therefore \boxed{S^* = T - \dfrac{1}{\lambda}}$ (ans).

(c) ~~[illegible]~~ $P(\text{win})(S^*) = \lambda\left(T - T + \dfrac{1}{\lambda}\right) e^{-\lambda\left(T - T + \frac{1}{\lambda}\right)}$

$= e^{-\lambda \cdot \frac{1}{\lambda}} = \dfrac{1}{e}$.

$\therefore$ Probability of winning for "optimal" 's'

is $\boxed{P(\text{win}) = 1/e}$

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3. → Infection process $N(t) \sim \text{Poisson}(\lambda t)$.
→ Detection delay $L \sim \text{Exp}(\mu)$.

∴ Detection = Infection + Delay

(a) P(detected by t | s) → Fixing infection time 's'.

∴ $s + L \leq t$
⇒ $L \leq t - s$.

∴ Since $L \sim \exp(\mu)$.

∴ $P(L \leq t - s) = 1 - e^{-\mu(t-s)}$.

∴ $\boxed{p(s) = 1 - e^{-\mu(t-s)}}$ (ans)

* early detection → large $(t-s)$ ⟨detection ↑⟩
* late detection → small $(t-s)$ ⟨missing ↑⟩

(b) Arbitrary infection is located UNIFORMLY on the interval $[0, t]$.

$$\therefore p = E\left[1 - e^{-\mu(t-s)}\right].$$

For uniform dist:

$$p = \frac{1}{t}\int_0^t \left[1 - e^{-\mu(t-s)}\right] ds \qquad e^{-\mu t} \int e^{\mu s} ds$$

$$= \frac{1}{t}\left[(t)_0^t - \int_0^t ds\, e^{-\mu(t-s)}\right]$$

$$= 1 - \frac{1}{t} \cdot \frac{e^{-\mu t}}{\mu}\left[e^{\mu t} - 1\right].$$

$$\boxed{p = 1 - \left(\frac{1 - e^{-\mu t}}{\mu t}\right)} \text{ (ans)}$$

(c) estimating λ.

$N_1(t) \longrightarrow$ detection $\sim$ Poisson $(\lambda t p)$.

$N_2(t) \longrightarrow$ NOT detecting $\sim$ Poisson $(\lambda t(1-p))$.

$E[N_1(t)] = \lambda t p.$ {detection}.

For estima^n we can $\{E[N_1(t)] = N_1(t)\}$

$\therefore N_1(t) = \hat{\lambda} t p.$

$$\hat{\lambda} = \frac{N_1(t)}{tp} = \frac{N_1(t)}{t\left[1 - \frac{1-e^{-\mu t}}{\mu t}\right]}.$$

$$\boxed{\hat{\lambda} = \frac{\mu t\, N_1(t)}{\mu t^2 - 1 + e^{-\mu t}}}$$ (ans).

(d) we want $E[N_2(t)]$. ~~[illegible]~~

$E[N_2(t)] = \lambda t(1-p)$

$E[N_2(t)] \approx \hat{\lambda} t(1-p)$

$$\therefore E[N_2(t)] = \frac{N_1(t)}{tp} \cdot t(1-p) = \boxed{N_1\left(\frac{1-p}{p}\right)}$$ (ans)

$$\boxed{p = 1 - \frac{1-e^{-\mu t}}{\mu t}}$$

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(4) Let : $S(t) = \sum_{i=1}^{N(t)} X_i$

where $N(t)$ is a poisson process of rate λ, independent of the iid sequence (X_i), with

$E[X_i] = \mu$, $Var(X_i) = \sigma^2$.

$Cov(N, S) = E[NS] - E[N] \cdot E[S]$

Our requirement is $Cov(N(t), S(t))$.

$E[N(t)] = \lambda t$.

By wald's identity : $E[S(t)] = \underbrace{E[N(t)] \cdot E[X]}_{= \lambda t \mu}$

(4) $S(t) = N(t)(\mu) + \text{noise}$ → deterministic [given N]; noise → fluctuations of X_i.

→ To calculate $E[N(t) \cdot S(t)]$,

$S(t) = \sum_1^n X_i \Rightarrow E[S(t) \mid N=n] = n \cdot \mu$.

$\therefore E[N(t) \cdot S(t) \mid N=n] = n \cdot n\mu = n^2 \mu$.

$\Rightarrow E[N \cdot S] = \mu \cdot E[N^2]$.

For poisson rv: $E[N^2] = Var(N) + (E[N])^2 = \lambda t + (\lambda t)^2$.

$\therefore E[N \cdot S] = \mu \lambda t(1 + \lambda t)$.

$\therefore Cov(N, S) = -\mu(\lambda t)^2 + \mu\lambda t(1 + \lambda t)$.

$= \mu\lambda t[(1 + \cancel{\lambda t}) - \cancel{(\lambda t)}] = \mu\lambda t$

$\therefore \boxed{Cov(N, S) = \mu\lambda t}$ (ans)

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(5) Emails arrive according to NONHOMOGENEOUS Poisson process with intensity $\lambda(t) = 5 + 2t$.

(a) For NHPP, mean measure on $[a,b]$ is:

$$m(a,b) = \int_a^b \lambda(u)\,du.$$

$\longrightarrow$ Mean on $[1,4] \Rightarrow \int_1^4 (5+2t)\,dt$

$$= 5(4-1) + (4+1)(4-1)$$

$$E[N(4) - N(1)] = (4-1)[5+5] = 30$$

(b) Mean $= \int_2^3 (5+2t)\,dt$.

$$= 5(3-2) + (3+2)(3-2).$$

$$\mu = 5 + 5 = 10,$$

$\therefore \lambda' = 10.$

$$\therefore P(N = \lambda') = \text{Poisson}(\lambda') = \boxed{e^{-10} \cdot \frac{10^{10}}{10!}} \text{ (ans)}$$

(c) Intensity in $[0,5] \rightarrow 5\cdot(5-0) + (5+0)(5-0)$.

$$\lambda_T = (50).$$

Intensity in $[3,5] = 5\cdot(5-3) + (5-3)(5+3)$

$$\lambda_t = 2\cdot(5+8) = 26$$

$$P\{N(5) - N(3)\} \mid \{N(5) - N(0) = 50\} = \text{Binomial}\left(50, \frac{\lambda_t}{\lambda_T}\right).$$

$\therefore P\left(N(5)-N(3) \geq 20 \mid N(5)-N(0)=50\right)$

$$= \sum_{k=20}^{50} \binom{50}{k} \cdot \left(\frac{26}{50}\right)^k \cdot \left(1-\frac{26}{50}\right)^{50-k} \text{ (ans.)}$$

6. Claims arrive as poisson process with rate $\lambda = 5/\text{day}$.

Claim amounts are i.i.d, independent of arrivals with $\mu = 100$, $\sigma^2 = 50^2 = 2500$.

Let $S(t) = \sum_{i=1}^{N(t)} X_i$.

(a) Since $N(t) \sim \text{Poisson}(5t)$

we have $E[N(t)] = 5t$ & $Var[N(t)] = 5t$.

claim size

$Var[S(t)] = E[N(t)] \cdot Var(X) + Var(N(t)) (E[X])^2$

$\rightarrow$ # of claims.

$E[S(t)] = E[N(t)] \, E[X]$.

$= (5t\mu) = (500t)$.

$\therefore Var[S(t)] = 5t \cdot 2500 + 5t \cdot 100^2$.

$= 5t \cdot 100 \cdot (25 + 100)$

$= 125 \times 100 \times 5t = 62500t$.

$\therefore$ $E[S(t)] = 500t$

$Var[S(t)] = 62500t$.

(b) Since $N(t) \sim \text{Poisson}(\lambda t)$.

$N(t) \sim \text{Poisson}(5t)$

$P(N(1) = 0) \sim \text{Poisson}(5)$

$\therefore P(\text{no claims on a day}) = e^{-5}$

(c) Given 10 claims, what is expected total.

If $N = 10 \longrightarrow S(t) \mid (N(t) = 10)$

$\therefore S = X_1 + X_2 + X_3 + \dots X_{10}$.

$\therefore E[S(t) \mid N = 10] = 10\mu$.

$= \boxed{1000}$.

(d) Expected capital @ $t = 30$ days.

$C(t) = 10000\, e^{0.05t} - S(t)$.

$E[C(t)] = 10000 \exp(.05t) - E[S]$.

@ $t = 30$

$E[S(30)] = 500 \times 30 = 15000$.

$\therefore E[C(30)] = 10000 \exp(.05 \times 30) - 15000$

$= 10000 \exp(1.5) - 15000$.

$\approx 2.98 \times 10^4 \approx \boxed{3 \times 10^4}$

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(7) $\Lambda \begin{cases} 2 \text{ claims/ day} \to p = 1/2 \\ 6 \text{ claims/ day} \to p = 1/2 \end{cases}$

(a) $P(\Lambda = 2) = \frac{1}{2}$, $P(\Lambda = 6) = \frac{1}{2}$.

$P(N(t) \mid \Lambda = \lambda) \sim \text{Poisson}(\lambda t)$.

$E[N(t)] = E[E[N(t) \mid \Lambda]] = E[\Lambda t] = t E[\Lambda]$.

$E[N(t)] = t\left\{\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 6\right\} = 4t$.

$\therefore E[N(t)] = 4t$.

$Var(N) = E[Var(N \mid \Lambda)] + Var(E[N \mid \Lambda])$.

Now, $Var(N \mid \Lambda = \lambda) = \lambda t$

$\Rightarrow E[Var(N \mid \Lambda = \lambda)] = E[\Lambda t] = 4t$.

Again,

$E[N \mid \lambda] = \lambda t$.

$\Rightarrow Var[E[N \mid \lambda]] = Var(\lambda t) = t^2 Var(\Lambda)$.

$= t^2 \{E[\lambda^2] - (E[\lambda])^2\}$.

$= t^2 \left\{\frac{1}{2} \cdot 2^2 + \frac{1}{2} \cdot 6^2 - \left(\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 6\right)^2\right\}$

$= t^2 \{2 + 18 - (4)^2\} = 4t^2$

$\therefore Var[N(t)] = 4t + 4t^2 = 4t(t+1)$ (ans)

(b) $P(N(4)=0) = \frac{1}{2} e^{-2} + \frac{1}{2} e^{-6}$

$P(\text{No claims}) = \approx 0.0689 \approx \boxed{0.07}$.

(c) $P(\Lambda=6 \mid N(1)=4)$

$= \dfrac{P(\Lambda=6) \cdot P(N(1)=4 \mid \Lambda=6)}{P(N(1)=4)}$

$P(N(1)=4 \mid \Lambda=6) = e^{-6} \cdot \frac{6^4}{4!}$,

$P(N(1)=4) = P(N(1)=4 \mid \Lambda=4) P(\Lambda=4)$
$\quad + P(N(1)=4 \mid \Lambda=6) P(\Lambda=6)$.

$P(N(1)=4) = \frac{1}{2} e^{-6} \cdot \frac{6^4}{4!} + \frac{1}{2} e^{-2} \cdot \frac{2^4}{4!}$,

$$P(\Lambda=6 \mid N(1)=4) = \frac{\frac{1}{2} e^{-6} \cdot \frac{6^4}{4!}}{\frac{1}{2} e^{-6} \cdot \frac{6^4}{4!} + \frac{1}{2} e^{-2} \cdot \frac{2^4}{4!}}$$

$$\therefore P(\Lambda=6 \mid N(1)=4) = \frac{e^{-6} \cdot 6^4}{e^{-6} \cdot 6^4 + e^{-2} \cdot 2^4},$$

$= 0.59735$. (ans).

(a) $E[\Lambda \mid N(1)=4]$.

Since Λ only takes 2/6.

$$E[\Lambda \mid N(1)=4] = 2\cdot P(\Lambda=2 \mid N(1)=4) + 6\cdot P(\Lambda=2 \mid N(1)=4).$$

let $P(\Lambda=2 \mid N(1)=4) = r$.

$\therefore P(\Lambda=6 \mid N(1)=4) = 1-r$.

$$\therefore E[\Lambda \mid N(1)=4] = 2r+6(1-r)$$
$$= 2r+6-6r$$
$$= 6-4r.$$

Now we found $r = 0.59735$ in (7c).

$$\therefore E[\Lambda \mid N(1)=4] = 6-4\times(0.59735)$$
$$= \boxed{3.610586} \text{ (ans.)}$$